

4	(a)	The vibration of the lattice ions increased, [B1] thus increasing the collisions between the free electrons and the lattice ions. [B1] This increases the resistance of the tungsten wire.	
	(b)	(i)	2075 °C [B1]
		(ii)	$P = \frac{V^2}{R}$ $R = \frac{V^2}{P}$ $= \frac{120^2}{60} \quad [C1]$ $= 240 \, \Omega \quad [A1]$
		(iii)	$R = \rho \frac{l}{A}$ $l = \frac{RA}{\rho}$ $= \frac{(240) \left(\pi \left(\frac{0.046 \times 10^{-3}}{2} \right)^2 \right)}{6.9 \times 10^{-7}} \quad [C1]$ $= 0.578 \, \text{m} \quad [A1]$
		(iv)	Length of filament can be short so that the <u>lamp can be made compact</u> OR tungsten wire can <u>fit into the lamp</u> .[B1]
		(v)	Since the internal resistance is of same resistance to the lamp, 50% of the power from the e.m.f. source is dissipated in lamp. 5.0% of his power dissipated in lamp is useful (in producing visible light. Hence efficiency $= (0.05)(0.50) \times 100\% \quad [C1]$ $= 2.5\% \quad [A1]$ Alternative answer. useful power = 5.0% x 60 = 3W input power = $I^2 R_{\text{eff}} = (0.50)^2 (240+240) = 120 \, \text{W}$ Hence, efficiency = $3/120 \times 100\% = 2.5 \%$

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5	(a)		
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